

B1GMB42 ZFS Recovery

B1GMB42 ZFS Recovery — rpool and bpool from live media

Machine: Dell Precision T5810 (Tower5810)

Installed pools: rpool (root + /home) on Hitachi sdb4 + TEAM sda special vdev; bpool (/boot) on Hitachi sdb2

Boot environment: rpool/R00T/ubuntu_cortt9, bpool/B00T/ubuntu_cortt9

Encryption: off (no ZFS passphrase on these pools)

Use this when the installed Ubuntu system will not boot. Boot **Ventoy Ubuntu 26.04** from **Wiggly** (sdc1) instead — do not import rpool until you are ready to repair.

Required host setting: force import

On the **installed** system, /etc/default/zfs must include:

```
ZPOOL_IMPORT_OPTS="-f"
```

Without this, boot can fail or hang after a recovery `zpool export`, `hostid` mismatch, or unclean shutdown — `initramfs import` will not force-import the pools. Recovery scripts already pass `-f` on the command line; the installed OS does **not** unless this default is set.

```
# verify on Tower5810 (or inside chroot after mount)
```

```
grep '^ZPOOL_IMPORT_OPTS' /etc/default/zfs
```

```
# expect: ZPOOL_IMPORT_OPTS="-f"
```

One-shot alternative: add `zfsforce=1` on the GRUB/kernel command line for a single boot.

1. Choose your path

Situation	What to use
Ventoy live, IndianaDell or DOSBOOT scripts available	Section 2 (recommended)
Ventoy live, no scripts — only a shell	Section 3 (manual zpool)
Installed system still boots but you need a read-only view	<code>mount-rpool-recovery.sh mount --overlay</code>

DOSBOOT copy: IndianaDell/recovery/ on partition **DOSBOOT** (sdc3, vfat).

Repo copy: ~/Documents/IndianaDell/mount-rpool-recovery.sh and scripts/recovery/.

2. With IndianaDell / DOSBOOT scripts

Prerequisites (live session)

```
sudo apt-get update
sudo apt-get install -y zfsutils-linux
```

Mount rpool for chroot (default)

```
cd /path/to/recovery/scripts      # DOSBOOT/IndianaDell/recovery or IndianaDell repo
sudo bash mount-rpool-recovery.sh mount      # use bash on DOSBOOT (vfat)
sudo bash mount-bpool-recovery.sh mount      # puts /boot at /recovery/boot
sudo bash mount-rpool-recovery.sh status
```

You should see /recovery/bin, /recovery/home, /recovery/etc, and /recovery/boot (kernel + initrd).

Enter chroot and repair

```
sudo ./mount-rpool-recovery.sh chroot
# inside chroot:
mount | grep boot
zpool status
apt-get update
apt-get install -y --reinstall zfs-initramfs grub-efi-amd64-signed shim-signed
update-initramfs -c -k all
update-grub
exit
```

Tear down

```
sudo ./mount-bpool-recovery.sh umount
sudo ./mount-rpool-recovery.sh umount
```

Overlay fallback (running system already on rpool)

Only when a full live-media chroot is impossible:

```
sudo ./mount-rpool-recovery.sh mount --overlay
ls /recovery/home/user
sudo ./mount-rpool-recovery.sh umount
```

3. Without scripts (manual commands)

Run as **root** from Ventoy Ubuntu live. Replace dataset names if yours differ (zpool import / zfs list to discover).

3a. Import and mount rpool

```
sudo mkdir -p /recovery
sudo zpool import -N -f -R /recovery -d /dev/disk/by-id rpool
sudo zfs mount -a -R /recovery
sudo zfs list -r rpool | head -20
```

Boot environment should appear under /recovery (e.g. /recovery/home/user).

3b. Import and mount bpool at /recovery/boot

```
sudo zpool import -N -f -d /dev/disk/by-id bpool
BOOT_DS=$(zfs list -H -o name,mountpoint -r bpool/B00T | awk '$2=="/boot"{print $1}')
sudo zfs set mountpoint=/recovery/boot "$BOOT_DS"
sudo zfs mount "$BOOT_DS"
ls /recovery/boot
```

3c. Chroot

```
for d in dev proc sys run; do sudo mount --bind /$d /recovery/$d; done
sudo mount --bind /dev/pts /recovery/dev/pts
sudo chroot /recovery /bin/bash
```

Inside chroot: run repairs (Section 4), then exit.

3d. Unmount and export

```
sudo zfs umount -a -R /recovery
sudo zpool set altroot=- rpool
sudo zpool export rpool
sudo zfs umount "$BOOT_DS"
sudo zpool export bpool
```

4. Common repairs (inside chroot)

Problem	Commands
Broken initramfs / boot	update-initramfs -c -k all && update-grub
Boot hangs on ZFS import after recovery	Ensure /etc/default/zfs has ZPOOL_IMPORT_OPTS="-f"; rebuild initramfs if needed
Pool won't import (live)	zpool import -f -d /dev/disk/by-id (inspect candidates); check dmesg
Pool degraded	zpool status -v; replace faulted vdev per ZFS docs
PERC H710 fault	SAS bays unavailable — see hardware manual; pool on SATA only
Boot menu missing	apt-get install --reinstall grub-efi-amd64-signed shim-signed
ZFS mount fails	zfs mount -a; check zfs list -o name,mountpoint,canmount

Before leaving chroot, confirm force import is set:

```
grep '^ZPOOL_IMPORT_OPTS' /etc/default/zfs
# must show: ZPOOL_IMPORT_OPTS="-f"
# if missing or empty:
#   printf 'ZPOOL_IMPORT_OPTS="-f"\n' | tee -a /etc/default/zfs # or edit in place
# then: update-initramfs -c -k all
```

Do not re-enable ZFS encryption until TPM + recovery are documented (hardware manual).

5. Disk map (this machine)

Device	Pool / role
sdb4	rpool main vdev
sda	rpool special vdev
sdb2	bpool
sdb1	EFI (/boot/efi when running)
sdcl Wiggly	Ventoy + Ubuntu live ISO + persistence
sdcl DOSBOOT	Recovery scripts + this manual

6. Script reference

Script	Purpose
mount-rpool-recovery.sh	Import rpool at /recovery, chroot, overlay, umount
mount-bpool-recovery.sh	Import bpool, mount boot FS at /recovery/boot

Environment overrides: POOL_NAME, BPOOL_NAME, RECOVERY_ROOT.

7. Verify after repair

Reboot to installed disk (not Ventoy). Then:

```
grep '^ZPOOL_IMPORT_OPTS' /etc/default/zfs # must be "-f"
zpool status rpool bpool
zfs list -r rpool | head -15
ls /boot/grub
```

If boot fails again, repeat from Section 2 or 3. If import stalls at boot, boot Ventoy, set ZPOOL_IMPORT_OPTS="-f" in the chroot (Section 4), update-initramfs, and retry.

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